

Coordinated Energy Scheduling in Interconnected Power-Water-Gas Networks Under Network Failures

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Abstract—The rapid growth of interconnections between different energy systems has seen emerging developments in recent years. Electric Power System (EPS), Water System (WS), and Natural Gas System (NGS) have been traditionally operated independently. However, a multi-sectoral approach is needed to enhance the resilience of critical infrastructure systems and reap the economic benefits. This becomes crucial as natural disasters with massive power outages have increased in both frequency and magnitude. To address these challenges, this paper proposes a cooperative joint framework for the operation of the Power, Water, and Gas Networks (PWGNs) during emergency operating conditions. The proposed formulation takes into account the nonlinear nature of both the hydraulic constraints in WS and the gas flow in the NGS. To reduce the computational burden of the original problem, linearization techniques are employed to convert the nonlinear model into a Mixed-Integer Linear Programming (MILP) formulation. Case studies with different contingency scenarios are analyzed in the IEEE 24-bus and IEEE 118-bus test systems, which are jointly operated with a 12-node and 20-node NGS, and a 15-node commercial-scale WS. Numerical results demonstrate that the proposed model effectively enhances the resilience of the interconnected energy systems and reduces the overall operational costs during emergency conditions. Additionally, results show that the proposed joint operation model provides significant energy flexibility and is efficiently scalable to large networks.

Index Terms—Contingency analysis, electric power system, water system, natural gas system, interdependency.

NOMENCLATURE

A. Sets

- $i \in \mathbf{I}$ Set of all dispatchable power plants.
 $c \in \mathbf{C}$ Non-natural gas dispatchable power plants.

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- $g \in \mathbf{G}$ Subset of natural gas-fired power plants.
 $j \in \mathbf{J}$ Set of wind power units.
 $t \in \mathbf{T}$ Set of time intervals.
 $l \in \mathbf{L}$ Set of power transmission lines (n, r) .
 $n \in \mathbf{N}$ Set of power network nodes (substations).
 $k \in \mathbf{K}$ Set of natural gas supply units.
 $z \in \mathbf{Z}$ Set of natural gas pipelines (m, u) .
 $m \in \mathbf{M}$ Set of natural gas network nodes.
 $v \in \mathbf{V}$ Set of fixed gas pressure points.
 $p \in \mathbf{P}$ Set of water system pumps.
 $s \in \mathbf{S}$ Set of water system pipes.

B. Parameters and Constants

- $D_{n,t}^E$ Electricity demand at node n at time t (MW).
 C_i^E Production cost of generating unit i (\$/MWh).
 C_t^{sh} The cost of load shedding in period t (MW).
 $P_i^{\max}$ Capacity of dispatchable power plant i (MW).
 $W_{j,t}$ Wind power forecast for unit j in period t (MW).
 $B_{n,r}$ Susceptance of transmission line (n, r) (S).
 $F_{n,r}^{\max}$ Capacity of transmission line (n, r) (MW).
 $D_{m,t}^G$ Natural gas demand at node m at time t (Kcf/h).
 C_k^G Supply cost of gas supply unit k (\$/Kcf).
 ϕ_i Power conversion factor of natural gas unit $i \in \mathbf{G}$ (kcf/MWh).
 $G_k^{\min, \max}$ Capacity of natural gas unit k (Kcf).
 $K_{m,u}$ Natural gas flow constant of pipeline (m, u) (kcf/psig).
 $S_{m,u}$ Linepack constant of pipeline (m, u) [kcf/(psig h)].
 $H_{m,u}^0$ Initial linepack in pipeline (m, u) (kcf).
 $PR_m^{\max}$ Maximum pressure limit at node m (psig).
 $PR_m^{\min}$ Minimum pressure limit at node m (psig).
 $\Gamma_{m,u}$ Compressor factor at gas pipeline (m, u) .
 $\mathcal{C}$ Reservoir-Node $\mathcal{R} \times \mathcal{N}$ incidence matrix.
 $\mathcal{K}$ Pump-Arc $\mathcal{P} \times \mathcal{A}$ incidence matrix.
 $\mathcal{G}$ Tank-Node $\mathcal{T} \times \mathcal{N}$ incidence matrix.
 Θ Bus-Pump $\mathcal{B} \times \mathcal{P}$ incidence matrix.
 d_t Vector of water demand (m^3/sec) at time t .
 $\hat{h}$ Reservoirs' geographical height (m).
 $q_{(\cdot)}^{\epsilon}, q_{(\cdot)}^{\hat{p}}$ Linearized flow rate for pipe ϵ and pump p .
 $V^{\max}, V^{\min}$ Maximum/minimum volume of water tanks.
 $\Delta E^{\max/\min}$ Maximum/minimum of the difference in the water flow rate to water tanks.
 $h^{\max}, h^{\min}$ Maximum/minimum nodal pressure heads.

$P_{p,\max/\min}$	Maximum/minimum power consumption of water pump p at each bus b (MW).
$z^{\max/\min}$	Maximum/minimum water flow rate in the network pipes (m^3/h)
M	A sufficiently large constant.

C. Decision Variables

$P_{i,t}, w_{j,t}$	Dispatch of units i and j at time t (MW).
$\theta_{n,t}$	Voltage angle at node n at time t (rad).
$f_{n,r,t}$	Power flow in line (n, r) in period t (MW).
$P_{n,t}^{sh}$	Power shedding in node (n) in period t (MW).
$g_{k,t}$	Dispatch of units k and j at time t (Kcf/h).
$pr_{m,t}$	Pressure at node m at time t (psig).
$q_{m,u,t}^{in/out}$	Inflow/outflow natural gas rate of pipeline (m, u) at time t (kcf/h).
$q_{m,u,t}$	Natural gas flow in pipeline (m, u) at time t (kcf/h).
R_t^s	Vector of water inflow rate from reservoir s at time t .
q_t	Water flow rate at time t .
h_t	Pressure heads loss for node n at time t .
$Sign(\cdot)$	Sign function applied during linearization.
ΔE_t	The difference in water tanks' inflow/outflow rate at time t .
$T_t^{in/out}$	Water inflow/outflow rates for tanks at time t .
V_t	Volume of stored water in tanks at time t .
W_t	Pumps' speed at time t .
P_t^p	Power consumption for pump p at time t .
$P^{p,wd}$	Vector of water electricity consumption wd for pump p at time t .
$X_{i,t}^\epsilon$	Continuous decision variable for the sampling coordinate i associated with the linearization of pipe ϵ at time t .
$X_{u,m,t}^p$	Continuous decision variable for the sampling coordinates u and m associated with the linearization of pump p at time t .
Kq_t	Water flow rate through pump at time t .
$a(\cdot), z(\cdot)$	Pump performance parameters.

D. Binary Variables

$v_{l,t}$	Binary variable on the connection status of power line l at time t (1 if online, 0 otherwise).
$y_{m,u,t}$	Binary variable defining the direction of the natural gas flow in pipeline (m, u) at time t .
$Y_{i,t}^\epsilon$	Binary variable for pressure head breakpoint i of water pipe ϵ at time t .
$h_{u,m,t}^{Upper,p}$	Binary variable for the upper left triangle in the linearization process of pump p at time t .
$h_{u,m,t}^{Lower,p}$	Binary variable for the lower left triangle in the linearization process of pump p at time t .

I. INTRODUCTION

POWER, water, and gas systems play a vital role in providing essential utility services to communities [1]. Community residents rely on the continuous delivery of electricity, gas, and water at all times. Traditionally, electricity, water, and gas

systems have been planned and operated independently of one another, while in reality, the operation of these systems is jointly interdependent [2], [3]. This interdependence is particularly evident in urban areas, where hydro systems, which contribute to a small percentage of the generated power, consume a substantial amount of energy for water pumping and distribution [4], [5]. Similarly, the interdependence between electricity and natural gas systems has become increasingly vital in recent years, as demonstrated by the significant increase in natural gas deliveries to electric power consumers in the US. This trend is exemplified by the power sector's growing share of total natural gas demand, which has risen from 22.3% to 35.5% over the last decade [6]. Therefore, treating each system independently might be inefficient as the occurrence of a failure, particularly in the case of High-Impact Low-Probability (HILP) events, in one system could easily cascade into the other systems resulting in catastrophic consequences to the society [7], [8], [9]. Further, besides considering the electrical grid, other infrastructures that are reliant on or impacted by the power grid must also be considered when assessing the resilience and the operational planning of the modern integrated energy infrastructure systems [10].

Given the challenges presented by the interdependence of power, water, and gas networks (PWGNs), a significant body of research has emerged in recent years that focuses on the optimization of these networks. The primary focus of these studies has been to minimize the total operational costs, enhance the system resilience, reduce greenhouse gas emissions and/or capture the dynamic behaviours of components in the WS. [11]. For the operation of the water and power systems (WPSs), [12] presented an uncertainty-informed coordinated day-ahead optimization model for integrated power and water networks considering the hydraulic constraints of the water system and the prevailing uncertainty in water demand and renewable energy sources. Reference [13] investigates the utilization of energy flexibility enabled through the coordination of power and water networks. Reference [14] conducted an economic dispatch analysis of water and power systems by considering the energy management of various building applications. A comprehensive study for the operational resilience of the interlinked water and power networks, considering the scenarios of limited availability of water and/or energy, is provided in [15]. Several metrics for determining the operational resilience of interdependent WPSs under critical conditions are discussed in [16]. In [17], a risk-based two-stage stochastic co-optimization framework that coordinates the operation of a renewable-rich power system with the operation of grid-connected water desalination plants is proposed to minimize operational costs while increasing the utilization of renewables.

As for the coupling between EPS and NGS, several works studied the optimal operation of power and natural gas systems under different conditions. In [18], the reliability assessment of the interconnected gas and power systems is explored considering power-to-gas and gas storage. Reference [19] constructs a robust scheduling model for wind-integrated energy systems that takes into account both gas pipeline and N-1 contingencies in power transmission. An analytical approach incorporating multiple variables for determining energy flow in natural gas

TABLE I
COMPARISON OF STUDIES ON INTEGRATED MULTI-ENERGY SYSTEMS

References	Integrated Networks	Model Type	Objective	Key Features
[12]	Power, Water	Dynamic	Co-optimization	MILP with JPC, water demand uncertainty
[15]	Power, Water	Static	Independent coordination	Genetic algorithm, drought and line outages
[21]	Power, Gas	Static	Cascading risk management	CC-MINLP, two-stage optimization
[22]	Power, Gas	Quasi-dynamic	Combined dispatch	CC-MILP, hurricane disruptions
[23]	Power, Water, Heat	Quasi-dynamic	Joint energy scheduling	MILP, sensitivity to heat and energy demand
[24]	Power, Gas, Water	Dynamic	Urban energy-water-carbon management	NCC-MILP, cascading outages
[25]	Power, Gas, Water	Static	Restoration planning	MINLP, hurricane resilience
Our Work	Power, Gas, Water	Dynamic	Emergency operation and resilience	MILP, transmission and pipeline failures

JPC: Joint Probabilistic Constraints, NCC: Nonlinear Conic Constraints, CC: Convex Constraints

networks is examined in [20]. A reliability assessment model for integrated electric power and natural gas systems is proposed in [21], which evaluates the performance of multiple components while considering cascading effects and renewable energy uncertainty. In [22], a two-stage dispatch model is formulated by first presenting a natural gas pipeline flow model that accounts for pipeline leakage and linepack, followed by a dispatch model that is considered to optimize the benefits for both electricity and natural gas suppliers.

However, for efficient and effective operation of the interconnected multi-energy sectors, it is crucial to incorporate the operation of water network in the comprehensive operational strategy. Given the interrelated nature of multi-energy systems, research has been undertaken to explore the development of optimal models for these interlinked networks. In [23], an integrated power-water-heat scheduling framework is introduced, emphasizing the sensitivity of system performance to variations in energy and heat demand. Similarly, [24] proposes the energy-water-carbon nexus framework by modeling interdependencies in power, gas, and water systems. Lastly, [25] develops a restoration-focused framework for power-water-gas distribution systems, employing resilience-motivated strategies to enhance system robustness and recovery capabilities post-disruption. Encouraged by the aforementioned reviewed studies, this paper extends the work in [26] by addressing a critical gap in the literature, which has predominantly focused on the operation of either power-water or power-gas networks, without fully capturing the interdependencies across all three interconnected networks. To bridge this knowledge gap, we propose an optimal operation strategy for integrated multi-energy networks under contingencies. Our model incorporates nonlinear hydraulic and gas flow constraints and evaluates system performance under line outage contingencies, enhancing system resilience and minimizing operational costs. Furthermore, the proposed approach demonstrates substantial energy flexibility and scalability, making it suitable for larger, more complex networks. Table I compares the contribution of this article to the representative previous studies. The main contributions of this paper are as follows:

- We propose a novel co-optimization strategy for power-water-gas networks (PWGNs), highlighting the benefits of integrating multiple energy carriers into a unified operational framework. This approach extends the existing studies by simultaneously addressing the interdependencies of all three networks and incorporating detailed network topologies.

- The proposed model incorporates advanced linearization techniques to address the nonlinear behaviors of hydraulic and gas flow systems. These methods enable computational efficiency while preserving modeling accuracy, making the framework suitable for large-scale systems.
- The resilience of the interconnected PWGNs is analyzed through detailed contingency scenarios, including transmission line and gas pipeline failures. The results demonstrate the framework's ability to mitigate cascading failures, minimize load shedding, and maintain operational stability under severe disruptions.

The rest of this paper is organized as follows: Section II introduces the proposed problem formulation for the PWGN's operation co-optimization. Section III explains the linearization technique applied to the proposed formulation. Section IV describes the numerical results, and Section V presents the concluding remarks.

II. PROBLEM FORMULATION

A. Proposed Framework

In this paper, we propose a day-ahead dispatch model for the coupled operation of multi-energy systems, i.e., PWGNs. While this is not yet approached in the current practice, the focus of this paper is to explore the potential benefits of such a cooperative approach economically and during emergency operations. To simplify the problem, we considered the DC model for the power network. Further, the cost associated with wind power generation is neglected. In the natural gas network, the gas flow dynamic equations can be approximated by accounting for the energy storage capacity in the form of linepack through variable inflows and outflows [27]. In the water network, the proposed model takes into account the hydraulic constraints in the WS [12], [28], [29]. The high-level inter-dependencies and coupling between power, water and gas networks are depicted in Fig. 1 and Fig. 2.

B. Objective Function

This objective function of the proposed co-optimization model is to minimize the total operation cost of PWGNs under emergency operations over a time period as follows:

$$\min \sum_{t \in \mathbf{T}} \left(\sum_{i \in \mathbf{C}} C_i^E p_{i,t} + \sum_{i \in \mathbf{N}} C_i^{sh} p_{i,t}^{sh} + \sum_{k \in \mathbf{K}} C_k^G g_{k,t} \right)$$

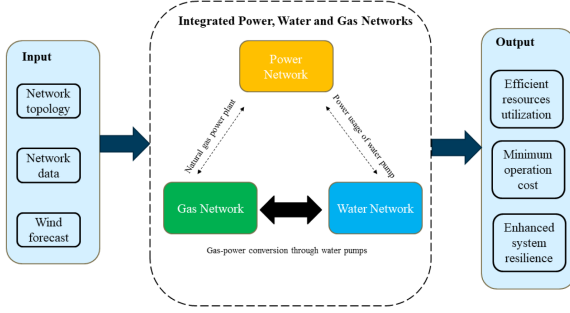


Fig. 1. The proposed power-water-gas nexus.

$$+ \sum_{s \in \mathbf{S}} C_{s,t} R_t^s + \sum_{p \in \mathbf{P}} C_{p,t} P_t^p \quad (1)$$

The first term in (1) represents the cost associated with the dispatchable power plants. The second term represents the cost of the load shedding at every node of the electricity network. The third term reflects the cost of supplying gas to natural gas-fired power plants. Water purchased from reservoirs is represented in the fourth term. Lastly, the cost of electricity needed to operate the water pumps is reflected in the fifth term. The objective function of the integrated PWGNs is subject to the following set of system and operation constraints.

C. Power System Model

Operation constraints for the power system's DC optimal power flow problem [3] and [27] are described in the following:

$$0 \leq p_{i,t} \leq P_i^{\max} \quad \forall i, \forall t \quad (2)$$

$$0 \leq w_{j,t} \leq W_{j,t} \quad \forall j, \forall t \quad (3)$$

$$f_{n,r,t} = B_{n,r}(\theta_{n,t} - \theta_{r,t}) \quad (n, r) \in \mathbf{L}, \forall t \quad (4)$$

$$-F_{n,r}^{\max} v_{l,t} \leq f_{n,r,t} \leq F_{n,r}^{\max} v_{l,t} \quad (n, r) \in \mathbf{L}, \forall t \quad (5)$$

$$-\pi \leq \theta_{n,t} \leq \pi \quad \forall n, \forall t \quad (6)$$

$$\sum_{i \in \mathbf{I}} p_{i,t} + \sum_{j \in \mathbf{J}} w_{j,t} - \sum_{(n,r) \in \mathbf{L}} f_{n,r,t} = D_{n,t}^E - \sum_{(n) \in \mathbf{N}} P_{n,t}^{sh} \quad \forall n, \forall t \quad (7)$$

Constraints (2) and (3) reflect the capacity limits of all power plants and wind farms, respectively. The power flows in transmission lines are characterized in (4) and the capacity of each line is restricted by its limits as in (5). In (6), the voltage angle at each node of the power network is restricted by its limits. Constraint (7) reflects the power balance at each node in the system.

D. Natural Gas System Model

The NGS schematic diagram and its components are illustrated in Fig. 3. The operational constraints of the NGS are

presented in (8)-(14) [30].

$$0 \leq g_{k,t} \leq G_k^{\max} \quad \forall g, \forall t \quad (8)$$

$$PR_m^{\min} \leq pr_{m,t} \leq PR_m^{\max} \quad \forall m, \forall t \quad (9)$$

$$pr_{u,t} \leq \Gamma_{m,u} \cdot pr_{m,t} \quad (m, u) \in \mathbf{Z}, \forall t \quad (10)$$

$$q_{m,u,t} \cdot |q_{m,u,t}| = K_{m,u}^2 \cdot (pr_{m,t}^2 - pr_{u,t}^2) \quad (m, u) \in \mathbf{Z}, \forall t \quad (11)$$

$$q_{m,u,t}^+ = \frac{q_{m,u,t}^{\text{in}} + q_{m,u,t}^{\text{out}}}{2} \quad (m, u) \in \mathbf{Z}, \forall t \quad (12)$$

$$q_{m,u,t}^- = \frac{q_{m,u,t}^{\text{in}} - q_{m,u,t}^{\text{out}}}{2} \quad (m, u) \in \mathbf{Z}, \forall t \quad (13)$$

$$\sum_{k \in \mathbf{K}} g_{k,t} - \sum_{i \in \mathbf{I}} \phi_i p_{i,t} - \sum_{(m,u) \in \mathbf{Z}} (q_{m,u,t}^{\text{in}} - q_{u,m,t}^{\text{out}}) = D_{m,t}^G \quad \forall (m, u) \in \mathbf{Z}, \forall t \quad (14)$$

Constraints (8) and (9) limit the capacity of gas supply and nodal pressure at each gas node, respectively. Gas pipelines with compressors relate the inlet and outlet pressures between the two adjacent nodes by the compressor factor Γ in (10). The piece-wise quadratic and non-convex constraint (11) models the Weymouth equation, which models the steady-state gas flow between two nodes. The flow in each gas pipeline can be modeled as the average of inflow and outflow gas as in (12) and (13). Finally, constraint (14) describes the nodal gas balance, which couples the power and natural gas systems through fuel consumption of Natural Gas Power Plants.

E. Water System Model

Fig. 4 illustrates the schematic diagram of the components in the WS. The mathematical formulation for the dynamic operation of the water network [31], taking into account water tank operational constraints, is modeled as follows:

$$R_t^s - d_t - \mathcal{M}^{-1} q_t - \mathcal{G} \Delta E_t = 0 \quad \forall t \quad (15)$$

$$\Delta E_t = T_t^{\text{in}} - T_t^{\text{out}} \quad \forall t \quad (16)$$

$$-q^{\max} \leq q_t \leq q^{\max} \quad \forall t \quad (17)$$

$$0 \leq \mathcal{K} q_t \leq q^{\max} \quad \forall t \quad (18)$$

$$R_t^s \geq 0 \quad \forall t. \quad (19)$$

$$-\mathcal{M} h_t = r_p |q_t|^{1.852} \text{Sign}(q_t) \quad \forall t \quad (20)$$

$$\mathcal{C} h_t - \hat{h} = 0 \quad \forall t \quad (21)$$

$$h^{\min} \leq h_t \leq h^{\max} \quad \forall t. \quad (22)$$

$$V_{t+1} = V_t + \Delta E_t \quad \forall t \quad (23)$$

$$V_t = S_a \mathcal{G} h_t \quad \forall t \quad (24)$$

$$V^{\min} \leq V_t \leq V^{\max} \quad \forall t \quad (25)$$

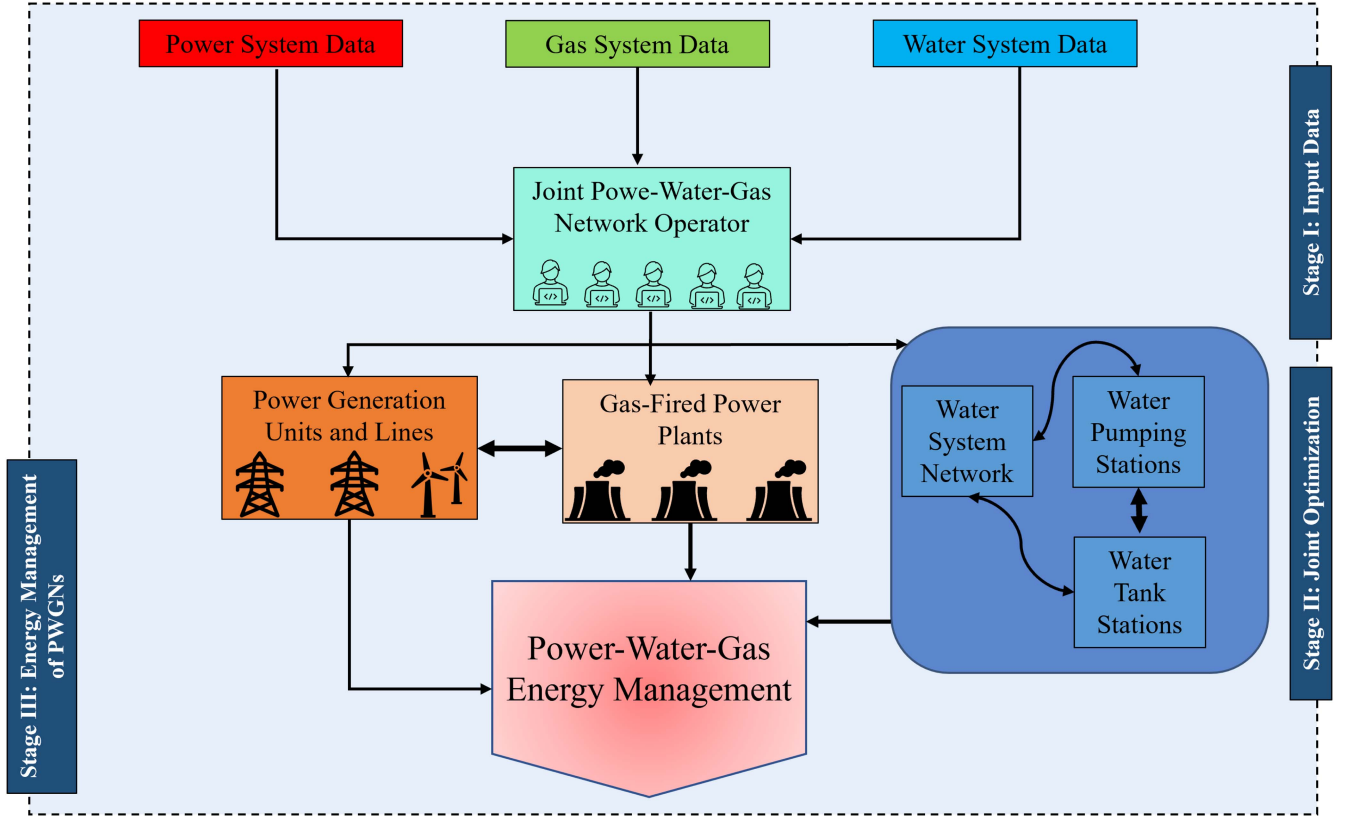


Fig. 2. The general proposed framework for joint optimization of Power-Water-Gas Operation.

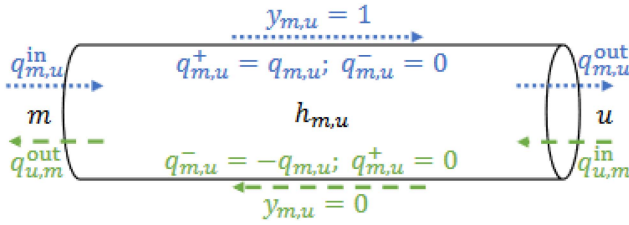


Fig. 3. Bidirectional Gas flow in a pip-line.

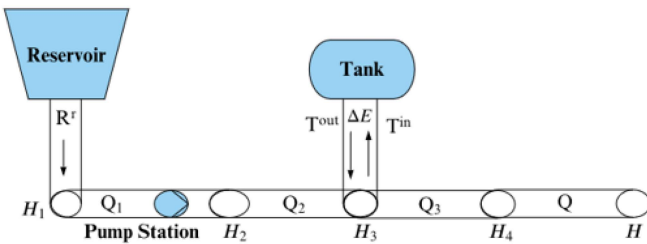


Fig. 4. Schematic diagram of components in water system.

$$P_t^p = W_t^3 \left(z_1 - z_2 \left(\frac{\mathcal{K}q_t}{W_t} \right) \right) \quad \forall t. \quad (29)$$

The balance in water flow is represented by constraint (15). The differences in water flow between the tanks' input and output are formulated by constraint (16). Constraint (17) limits the flow of water through pipes, while constraint (18) limits the flow of water through pumps. At reservoir nodes, constraint (19) restricts inflow rates to positive values. Constraints (20)-(22) represent the nodal pressure head loss for water pipes, which are modeled using the Hazen-Williams formula [32]. Modeling the dynamic operation of tanks and pumps, which are considered the most challenging components of a water network, is done through constraints (23)-(27). Constraint (28) controls the increase in water pump pressure, while constraint (29) models the electricity consumption required for water pumps. Further details on the operational constraints of the water network can be found in [12]. All water network constraints (15)-(29) appear as non-linear programming (NLP) constraints, where nonlinear constraints are presented in (20), (28), and (29).

$$\Delta E^{\min} \leq \Delta E_t \leq \Delta E^{\max} \quad \forall t \quad (26)$$

$$P^{p,\min} \leq P_t^p \leq P^{p,\max} \quad \forall t \quad (27)$$

$$\mathcal{K}\mathcal{M}h_t = W_t^2 \left(a_1 - a_2 \left(\frac{\mathcal{K}q_t}{W_t} \right)^{a_3} \right) \quad \forall t \quad (28)$$

III. LINEARIZATION TECHNIQUE

The joint coupling between EPS and NGS is reflected in (14), which couples the natural gas network with the power network through the natural gas-fired power plants. Similarly, the water network is coupled with the power network through the

electricity consumption of water pumps in (29). Since the resulting model is mixed-integer nonlinear programming (MINLP), which is computationally challenging to solve in large-scale systems, several linearization techniques are employed to transform the proposed MINLP model into a mixed-integer linear programming (MILP) formulation, enabling efficient computation with off-the-shelf solvers. For the water system, the nonlinear constraints, such as the Hazen-Williams equation for pipe flow and the pump electricity consumption relationship, are approximated using a piecewise linearization approach. This method ensures computational efficiency while preserving the accuracy of hydraulic dynamics. For the natural gas system, the nonlinear Weymouth equation is linearized using an outer approximation technique. This method generates a set of linear constraints based on Taylor series expansion around predefined pressure points, effectively capturing the nonlinear gas flow behavior. These techniques not only improve computational efficiency but also maintain the key operational characteristics of the water and gas systems. The following sections describe the linearization methods used for these two systems in detail, highlighting how the nonlinear constraints are addressed while ensuring the feasibility and scalability of the optimization framework.

A. Outer Approximation

In NGS, constraint (11) represents the Weymouth equation for the gas flow, which appears as a nonlinear constraint. In order to transform this constraint into a tractable mixed integer linear problem (MINP) constraint, we apply the outer approximation approach –based on the Taylor series expansion around fixed pressure points – for linearization [33]. The equality constraint in (11) is replaced by a set of the following linear inequalities:

$$q_{m,u,t} \leq K_{m,u} \left(\frac{PR_{m,v}}{\sqrt{PR_{m,v}^2 - PR_{u,v}^2}} pr_{m,t} - \frac{PR_{u,v}}{\sqrt{PR_{m,v}^2 - PR_{u,v}^2}} pr_{u,t} \right) \forall v, \forall t \quad (30)$$

V here refers to the set of fixed pressure points $PR_{m,v}, PR_{u,v}$. When we consider a significant number of fixed points (e.g., 20 points as in [33]), we can ensure a sufficient approximation of the Weymouth (11). Now, using the outer approximation method which is given by a number of planes in (30) that are tangent to the cone described by (11), the gas flow in each pipeline will be approximated by the only constraint in (30) that is binding. Hence, due to differing pressure limits between adjacent nodes, the flow for one direction in a pipeline may differ from the flow for the opposite direction. That's why in (30), the flow from node m to node u might be different from the flow from node u to node m . Accordingly, we introduce two non-negative variables $q_{m,u,t}^+, q_{m,u,t}^-$ and the binary variable $y_{m,u,t}$ to account for the bidirectional gas flow.

$$q_{m,u,t} = q_{m,u,t}^+ - q_{m,u,t}^- \quad (m, u) \in \mathbf{Z}, \forall t \quad (31)$$

$$0 \leq q_{m,u,t}^+ \leq M \cdot y_{m,u,t} \quad (m, u) \in \mathbf{Z}, \forall t \quad (32)$$

$$0 \leq q_{m,u,t}^- \leq M(1 - y_{m,u,t}) \quad (m, u) \in \mathbf{Z}, \forall t \quad (33)$$

The natural gas flow in each pipeline is decomposed in (31) to $q_{m,u,t}^+$ and $q_{m,u,t}^-$, which will determine the direction of flow using the binary variable y in (32) and (33). Note that the sufficiently large number M is set to the maximum gas flow along a pipeline to ensure that either (32) or (33) is valid at each time period.

Now, the non-linear constraint (11) is substituted by (34) and (35), which linearly approximate the Weymouth equation around the fixed pressure points.

$$q_{m,u,t}^+ \leq \frac{K_{m,u} PR_{m,v}}{\sqrt{PR_{m,v}^2 - PR_{u,v}^2}} pr_{m,t} - \frac{K_{m,u} PR_{u,v}}{\sqrt{PR_{m,v}^2 - PR_{u,v}^2}} pr_{u,t} + M(1 - y_{m,u,t}), \forall (m, u) \in \mathbf{Z} | (m > u), \forall v, \forall t \quad (34)$$

$$q_{m,u,t}^- \leq \frac{K_{m,u} PR_{u,v}}{\sqrt{PR_{u,v}^2 - PR_{m,v}^2}} pr_{u,t} - \frac{K_{m,u} PR_{m,v}}{\sqrt{PR_{u,v}^2 - PR_{m,v}^2}} pr_{m,t} + M y_{m,u,t} \forall (m, u) \in \mathbf{Z} | (u > m), \forall v, \forall t \quad (35)$$

Based on this, the direction of flow can be determined by the binary variable y . The value of y will determine whether (34) or (35) holds true.

B. Piece-Wise Linearization

In order to ensure a viable set of solutions for the optimization problem exists, a piece-wise linearization approach [34] is employed to handle the presented nonlinear constraints. The non-linear constraints (20), (28) and (29) are replaced by the following linear constraints:

$$\sum_{i=1}^{I-1} Y_{i,t}^\epsilon = 1 \quad \forall \epsilon, \forall t \quad (36)$$

$$X_{i,t}^\epsilon \leq Y_{i-1,t}^\epsilon + Y_{i,t}^\epsilon \quad \forall i = 2, \dots, I-1, \forall \epsilon, \forall t \quad (37)$$

$$\sum_{i=1}^I X_{i,t}^\epsilon = 1 \quad \forall \epsilon, \forall t \quad (38)$$

$$X_{I,t}^\epsilon \leq Y_{I-1,t}^\epsilon \quad \forall \epsilon, \forall t \quad (39)$$

$$X_{1,t}^\epsilon \leq Y_{1,t}^\epsilon \quad \forall \epsilon, \forall t \quad (40)$$

$$q_t^\epsilon = \sum_{i=1}^I X_{i,t}^\epsilon \hat{q}_i^\epsilon \quad \forall \epsilon, \forall t \quad (41)$$

$$\Delta h_t^\epsilon = \sum_{i=1}^I X_{i,t}^\epsilon \Delta h_i^\epsilon(\hat{q}_i^\epsilon) \quad \forall \epsilon, \forall t \quad (42)$$

$$\sum_{u=1}^U \sum_{m=1}^M X_{u,m,t}^p = 1 \quad \forall p, \forall t \quad (43)$$

$$\sum_{u=1}^U \sum_{m=1}^M (h_{u,m,t}^{Upper,p} + h_{u,m,t}^{Lower,p}) = 1 \quad \forall p, \forall t \quad (44)$$

$$X_{u,m,t}^p \leq h_{u,m-1,t}^{Upper,p} + h_{u+1,m,t}^{Upper,p} + h_{u,m,t}^{Upper,p} + h_{u-1,m,t}^{Lower,p} + h_{u,m+1,t}^{Lower,p} + h_{u,m,t}^{Lower,p} \quad \forall u, \forall m, \forall p, \forall t \quad (45)$$

$$q_t^p = \sum_{u=1}^U \sum_{m=1}^M X_{u,m,t}^p q_u^p \quad \forall p, \forall t \quad (46)$$

$$W_t = \sum_{u=1}^U \sum_{m=1}^M X_{u,m,t}^p w_m^p \quad \forall p, \forall t \quad (47)$$

$$\Delta h_t^p = \sum_{u=1}^U \sum_{m=1}^M \Delta h_t^p(q_u^p, w_m^p) X_{u,m,t}^p \quad \forall p, \forall t \quad (48)$$

$$P_t^p = \sum_{u=1}^U \sum_{m=1}^M P_t^p(q_u^p, w_m^p) X_{u,m,t}^p \quad \forall p, \forall t. \quad (49)$$

Constraint (36) ensures that only one binary variable $Y_{i,t}^\epsilon$ can take on a non-zero value. Constraints (37)-(40) dictate that only non-zero values are chosen for a pair of consecutive positive continuous variables $X_{i,t}^\epsilon$ and $X_{i+1,t}^\epsilon$. Constraint (41) can uniquely determine any given point for the water flow rate through pipes q_t^ϵ , as it is a linear combination of two successive breakpoints weighted by the associated variables $X_{i,t}^\epsilon$. Constraint (42) ensures that the pressure difference for each pipe Δh_t^ϵ is properly selected for the accurate computation of the approximated values.

The weights of the convex combination for the selected triangle are presented in (43). Equation (44) ensures that only a single triangle is utilized for the convex combination, while constraint (45) enforces that non-zero values of $X_{u,m,t}^p$ can only be associated with the three vertices of the triangle. Further, equations (46) and (47) represent linear combinations of any values for q_t^p and W_t , respectively, weighted by the continuous variables $X_{u,m,t}^p$. Finally, the bi-variate nonlinear functions for the increase in pressure difference in each water pump, represented by $\Delta h^p t$, and the power consumption of pumps, represented by P_t^p , are approximated in (48) and (49), respectively.

IV. NUMERICAL CASE STUDIES

A. Illustrative Test Systems

In this study, we considered three systems: a modified IEEE 24-bus Reliability Test System, a 12-node natural gas system and a 15-node commercial-scale water system. The power and gas system datasets are derived from [26], whereas the water system data is obtained from [12]. The projected profiles for power load, gas demand and wind power generation are depicted in Fig. 5. All optimization models are executed on a computer with an Intel 2.5 GHz Dual-Core processor with 16 GB of RAM, utilizing the GUROBI commercial solver.

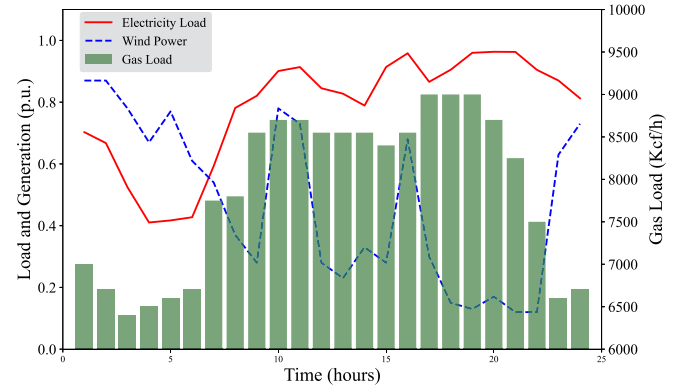


Fig. 5. The total power load, gas demand and wind power forecast: IEEE 24-bus test system.

TABLE II
OUTAGE SCENARIOS IN THE STUDIED CASES: IEEE 24-BUS TEST SYSTEM

Case Study	I	II	III	IV
Outage Scenario	None	L/11	Z/5	L/11 + Z/5

B. Normal and Emergency Operation

To evaluate the effectiveness of the proposed joint PWGNs operation model, four case studies are examined, with the corresponding outage scenarios summarized in Table II:

- *Case Study 1 (CSI) Normal Operating Conditions:* This case represents the baseline scenario with no outages, where the PWGNs operate under nominal conditions for a 24-hour period.
- *Case Study 2 (CSII) Line Tripping:* This case analyzes the impact of a single line tripping event in the power network.
- *Case Study 3 (CSIII) Gas Pipeline Failure:* Here, a critical gas pipeline failure is simulated, affecting fuel supply to gas-fired power plants.
- *Case Study 4 (CSIV) Combined Power and Gas Contingencies:* This scenario explores the simultaneous failure of a power transmission line and a gas pipeline.

These case studies considers both normal and emergency operating conditions under the N-1 security criterion. Specifically, we simulate the tripping of Line 11, which connects bus 7 to bus 8 in IEEE 24-Bus Test System, in Contingency Scenario 2 (CSII). Line 11 is critical as bus 7 is connected to the rest of the network through this line alone. An outage on Line 11 isolates bus 7, leading to an islanded operation. However, bus 7 is equipped with a wind farm, which can meet part of its local demand. The presence of the wind farm ensures that bus 7 can be self-sufficient under favorable wind conditions, thereby reducing the likelihood of cascading failures. It is worth noting that during normal operation (CSI), the power flow in each line is described by (4) and the system-wide power balance is described by (7). However, when Line 11 is tripped (CSII), bus 7 becomes isolated, where the power balance for bus 7 in such scenario is described as:

$$p_{G7,t} + w_{7,t} = D_{7,t}^E - P_{7,t}^{sh}, \quad \forall t \quad (50)$$

Where $p_{G7,t}$ is the power generated by conventional units at bus 7, and $w_{7,t}$ is the wind power generation at bus 7. For the remaining interconnected system (without bus 7), the power balance must account for the redistribution of power flows due to the tripping of Line 11. This redistribution is governed by *Line Interruption Distribution Factor (LIDF)* approach [35], which quantifies the impact of the outage on power flows across the rest of the network as in the following:

$$\Delta f_{l,t}^{\text{new}} = \Delta f_{l,t}^{\text{old}} + \sum_{k \in \mathbf{L}} \text{LIDF}_{l,k} \cdot \Delta f_{k,t}^{\text{outage}}, \quad \forall t, l \quad (51)$$

where:

- $\Delta f_{l,t}^{\text{new}}$ is the adjusted power flow on line l after the outage,
- $\Delta f_{l,t}^{\text{old}}$ is the original power flow on line l before the outage,
- $\text{LIDF}_{l,k}$ is the Line Interruption Distribution Factor between lines l and k ,
- $\Delta f_{k,t}^{\text{outage}}$ is the change in power flow on line k caused by the outage.

And for the rest of the interconnected system, the power balance is updated as follows:

$$\begin{aligned} \sum_{i \in \mathbf{I}} p_{i,t} + \sum_{j \in \mathbf{J}} w_{j,t} - \sum_{(n,r) \in \mathbf{L}} f_{n,r,t}^{\text{new}} \\ = D_{n,t}^E - \sum_{(n) \in \mathbf{N}} P_{n,t}^{\text{sh}}, \quad \forall n \neq 7, \forall t. \end{aligned} \quad (52)$$

This formulation ensures that the system remains balanced after the outage, with power being redistributed through the remaining transmission lines, and local generation at bus 7 managing the load at the islanded node.

In Contingency Scenario 3 (CSIII), we analyze the failure of gas pipeline 5, a critical supply line for gas-fired power plants. This outage substantially limits natural gas availability, impacting generation capacity. Further testing with failures in other pipelines highlighted the resilience limits of the gas network; specifically, an outage in pipeline 6 (linking node 4 to node 7) proved crucial. Pipeline 6 supplies gas storage, serves two gas-fired power plants, and meets direct demand at node 7. Its failure leads to infeasibility in the model, as the disruption creates a bottleneck that prevents the system from meeting both power and gas demands.

In Contingency Scenario 4 (CSIV), an outage of power line 11 and gas pipeline 5 creates a cascading effect, critically constraining both electricity generation and gas supply. With reduced gas-fired capacity and limited wind generation, load shedding becomes unavoidable in some areas. Note that no emergency operation is considered for the water network in this study. However, a sensitivity analysis will explore system scalability and resilience under increased water demand in a larger configuration.

C. Numerical Results and Discussions

In CSI, the electricity consumption by water pumps remains consistent, as shown in Fig. 6. The steady demand profile reflects normal operation, where water pumps function without additional strain from network disruptions. Fig. 7 presents the scheduled power generation of the five non-gas-fired power

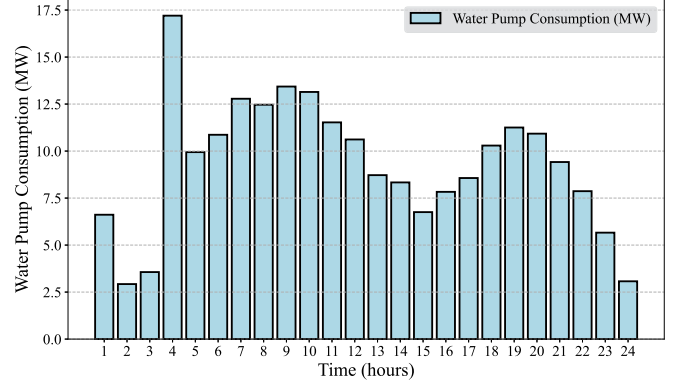


Fig. 6. The total electricity consumption of water pumps in CS1: IEEE 24-Bus Test System.

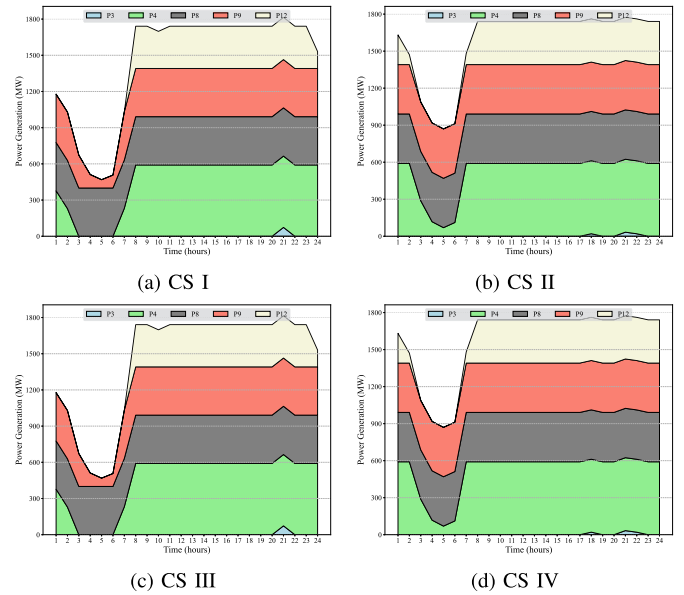


Fig. 7. Total power generation from non-gas fired power plants in IEEE 24-bus test system.

plants across all case studies. Notably, in CSI and CSIII, the dispatch of non-gas units remains relatively stable, indicating a balanced operation without substantial external demands. However, in CSII and CSIV, unit P12 demonstrates increased activity, particularly during the early hours, to compensate for the loss of wind production. This adjustment highlights the system's reliance on higher-priced power generation units to replace lost capacity due to network outages, particularly under conditions where wind generation is unavailable. Fig. 8 shows the total natural gas supply from Gas Well K3, which recorded the highest gas output across all scenarios in Table II. In response to contingencies in CSII, CSIII, and CSIV, the natural gas supply increases to compensate for the loss in generation capacity. This increase reflects the system's adaptive response, where gas supply ramps up to stabilize the network under emergency conditions, demonstrating the critical role of gas reserves in maintaining operational continuity.

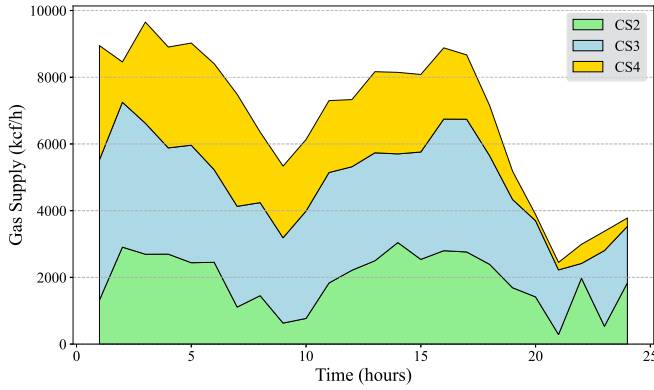


Fig. 8. The total natural gas supply of Unit K3: IEEE 24-bus test system.

TABLE III
OPERATION AND LOAD SHEDDING COSTS (K\$) AND COMPUTATION TIME (S)
FOR EACH CASE STUDY: IEEE-24-BUS TEST SYSTEM

Network type	CSI	CSII	CSIII	CSIV
Power	641.7	789.9	832.4.6	844.5
Water	404.4	385.7	378.6	361.9
Gas	525.9	672.6	589.4.2	781.7
Load Shedding Cost	0	87.4	42.1	100.8
Total Cost	1570.6	1856.7	1693.1	1962.4
Computation Time	866	1045	988	1083

During emergency operations, the spatial and temporal distribution of load shedding across nodes and time is shown in Fig. 9. In CSII (Fig. 9(a)), load shedding is concentrated around nodes 7, 8, 21, 22, and 23 during peak demand hours (18–22), when wind power generation is at its lowest. The failure of line 11 connecting nodes 7 and 8 worsen this condition, significantly impacting system stability. In CSIII (Fig. 9(b)), the failure of gas pipelines 5–6 limits gas supply to gas-fired power plants connected to nodes 22 and 23. As a result, load shedding is primarily observed at nodes 21, 22, and 23, with reduced severity compared to CSII due to the localized nature of the gas outage. The critical impact is still centered during hours 18–22, reflecting the system’s sensitivity to both gas supply and wind generation shortfalls. In CSIV (Fig. 9(c)), the simultaneous failure of line 11 and gas pipelines 5–6 leads to a more severe cascading effect, resulting in widespread load shedding across nodes 7, 8, 20, 21, 22, and 23. The load shedding is most observed during hours 18–22, highlighting the compounded vulnerabilities introduced by multi-network failures. Despite the broader distribution, the magnitude of curtailment at these critical nodes shows their importance in maintaining system reliability.

Table III presents the operation and load shedding costs for each case study, along with the computation times required to reach optimal solutions. Under normal conditions (CSI), the total operating cost is the lowest at \$1570.6k, with no load shedding cost incurred. The power, water, and gas networks contribute \$641.7k, \$404.4k, and \$525.9k, respectively, to this baseline cost. In CSII, the total cost rises to \$1856.7k, reflecting a \$286.1k increase from CSI due to a power line tripping event. This

increase is primarily driven by a load shedding cost of \$87.4k, highlighting the financial burden of maintaining system balance under disrupted conditions. The power network costs increase to \$789.9k, as higher-cost generation units are utilized. Conversely, the water network cost slightly decreases to \$385.7k, while the gas network cost rises to \$672.6k to accommodate the contingency. For CSIII, where a critical gas pipeline failure is simulated, the total cost is \$1693.1k, which is \$122.5k higher than CSI but \$163.6k lower than CSII. This scenario incurs a load shedding cost of \$42.1k, significantly less than CSII. The gas network cost is \$589.4k, reflecting the increased expense of securing alternative gas supplies. Despite the gas contingency, the power network cost remains high at \$832.4k due to the interconnected impact of the failure. These results demonstrate that gas pipeline failures can have localized impacts with relatively lower system-wide costs compared to power line failures. CSIV represents the most severe scenario, with both power and gas contingencies leading to the highest observed total cost of \$1962.4k—an increase of \$391.8k compared to CSI. The combined outages result in a load shedding cost of \$100.8k, the highest across all scenarios. The power network cost rises to \$844.5k, while the gas network cost reaches \$781.7k, reflecting the compounded strain on the integrated network. The computation time also increases across the scenarios, from 866 seconds in CSI to 1083 seconds in CSIV, reflecting the added complexity of solving the optimization problem under emergency conditions. It can be observed that the lower total costs and load shedding in CSIII compared to CSII demonstrate that gas contingencies, though impactful, may lead to less severe system-wide effects compared to certain power contingencies.

D. Sensitivity to Variations in Wind Capacity and Demand

The performance of multi-energy networks is highly influenced by uncertainties in renewable energy generation and electricity demand. To evaluate the robustness of the proposed framework under such variations, we employ Monte Carlo Simulation (MCS) to generate a diverse set of scenarios for wind power and electricity demand. Sensitivity analysis is conducted on these MCS-generated scenarios to systematically evaluate their operational impacts on cost, load shedding, and resilience metrics. Fig. 10 illustrates the objective function value as a function of (a) wind power penetration levels and (b) demand power penetration levels across the case studies CSII, CSIII, and CSIV. In Fig. 10(a), the costs increase as wind power penetration decreases, reflecting the rising reliance on higher-cost thermal generation to compensate for the reduced availability of wind power. For instance, in CSIV, the total cost increases from \$1.9M at 100% wind penetration to \$3.7M at 10% wind penetration, highlighting the compounded challenges posed by simultaneous power and gas contingencies. A similar trend is observed in CSII and CSIII, where the total costs rise from \$1.8M and \$1.7M at full wind penetration to \$3.6M and \$2.95M, respectively, at low wind levels. Notably, CSIII consistently demonstrates lower total costs compared to CSII due to the localized nature of the gas pipeline failure and its comparatively limited impact on the

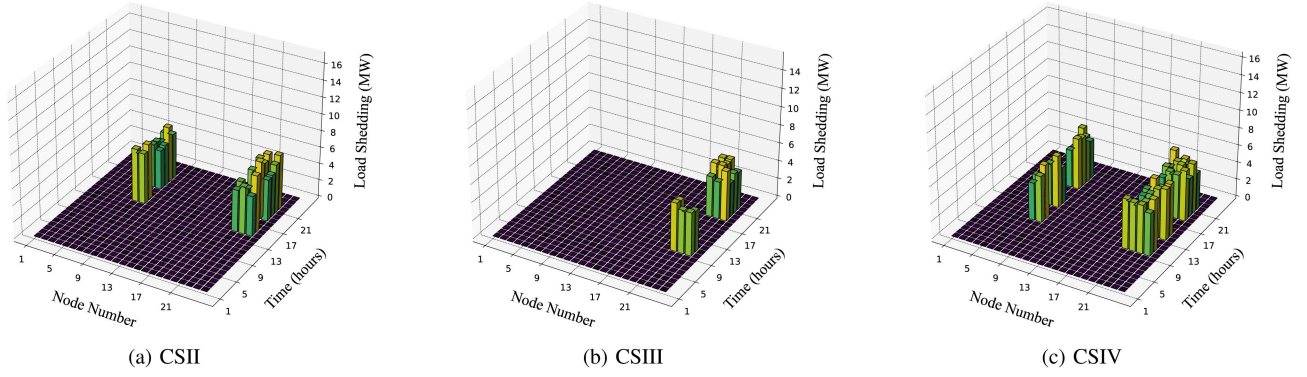


Fig. 9. Total spatial load shedding: IEEE 24-bus test system.

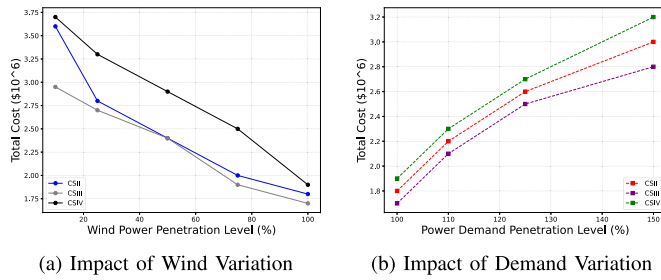


Fig. 10. Total cost of PWGNs as a function of (a) Wind Power Penetration Level (%) and (b) Power Demand Level (%) for CSII, CSIII, and CSIV: IEEE 24-bus test system.

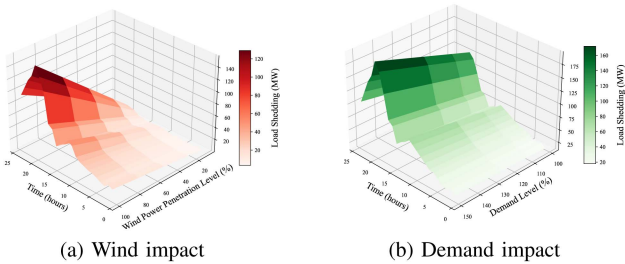


Fig. 11. Load shedding behavior in CSIV as a function of (a) wind penetration % and (b) power demand levels: IEEE 24-bus test system.

overall system. In contrast, Fig. 10(b) shows that as power demand penetration increases, total costs rise across all cases. For example, in CSIV, the total cost increases from \$1.9M at 100% demand penetration to \$3.2M at 150% demand penetration. Similarly, CSII and CSIII see increases from \$1.8M and \$1.7M to \$3.0M and \$2.8M, respectively. Fig. 11 illustrates the total load shedding in CSIV across different wind penetration (Figure (a)) and demand levels (Figure (b)). In Figure (a), reduced wind penetration correlates with increased load shedding, particularly during peak demand hours, showing the reliance on wind resources to balance supply and demand in contingency scenarios. As wind availability decreases, alternative generation sources must increase their output to meet demand, amplifying load curtailment needs when these resources are limited. Similarly,

TABLE IV
RESULTS WITH SEVERAL INTERCONNECTED WS: IEEE 118-BUS TEST SYSTEM AND MODIFIED BELGIAN 20-NODE GAS SYSTEM

Case Study	CS1	CS2	CS3	CS4
Wind Level	100%	90%	80%	70%
Cost (k\$)				
Power	133.3	17531.1	1858.9	2026.8
Gas	1071.8	1282.9	943.3	944.7
Water	1213.2	1199.1	1186.8	1167.9
Load Shed (MW)	0	49.7	17.2	359.5
Total Cost (k\$)	3681.3	4284.8	4006.1	4498.7
Computation time (s)	1011	1268	1094	1466

Figure (b) shows the load shedding required to maintain system stability under varying demand levels. Higher demand necessitates greater load curtailment during peak periods, indicating the system's vulnerability to demand surges, especially under emergency conditions where resource availability is constrained.

E. Scalability and System Resilience With Larger Integrated Networks

In this section, we examine the scalability of the proposed model by applying it to a larger system. Specifically, we test a modified IEEE 118-Bus Test System interconnected with a 20-Node Gas System and three 15-Node Water Networks. The interconnected test system includes 46 NGPP units, 8 GFPP units, 5 wind farms located at buses 2, 33, 51, 81, and 108. Additionally, there are 186 power lines, 91 electrical loads, 2 gas suppliers, 17 gas pipelines, and 9 gas loads, along with the three water networks connected to the power system at buses 39, 74, and 102. Detailed data can be referenced from [12] and [36]. Other parameters are consistent with those used in the IEEE 24-bus RTS.

The same four contingency cases (CSI–CSIV) studied in the IEEE 24-bus RTS are analyzed in this expanded network to evaluate both the operational cost and resilience under varying wind penetration levels. As shown in Table IV, the total costs and load shedding vary significantly across the cases, reflecting the increased complexity and interdependencies of the PWGNs. The cases were evaluated at varying wind penetration levels to assess the impact of wind integration on system performance. In CSII,

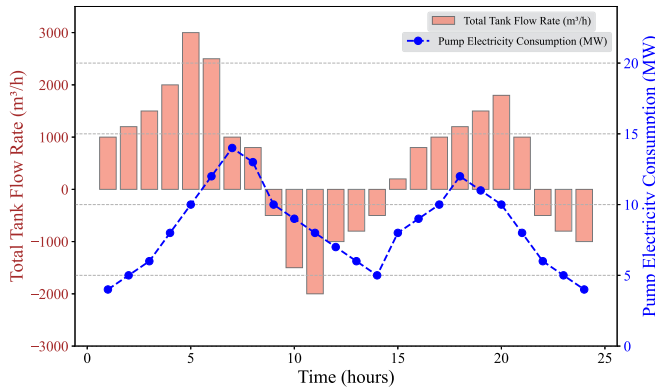


Fig. 12. Total tank flow rate and pump electricity consumption of water networks in CSIV: IEEE 118-bus test system.

a line outage occurs at line 150, connecting nodes 94 and 96. In CSIII, failures are observed in gas pipelines 10, which connect pipelines 12 and 13. CSIV combines these two contingencies, leading to the highest levels of operational disruption. Under normal operation (CSI), the system incurs a baseline total cost of \$3,681.3k with no load shedding. However, contingency cases lead to substantial cost increases, with CSIV resulting in the highest total cost of \$4,498.7k due to combined power and gas disruptions and a significant load shedding of 359.5 MW. CSII, which involves a single power line failure and reduced wind penetration, incurs a total cost of \$4,284.8k and a load shedding of 49.7 MW. In CSIII, where a critical gas pipeline failure occurs, the total cost is slightly lower at \$4,006.1k, with a corresponding load shedding of 17.2 MW. The operational costs across power, gas, and water networks reveal how these interconnected systems respond under different contingencies. The integration of additional water networks and an expanded gas network amplifies the total costs compared to the smaller IEEE 24-bus Test System, highlighting the increased interdependencies and vulnerabilities of larger networks. Furthermore, the computation time increases from 1,011 seconds in CSI to 1,466 seconds in CSIV, demonstrating the scalability of the proposed model to solve larger, more complex systems efficiently.

Fig. 12 illustrates the total water tank flow rate and pump electricity consumption for each water network during CSIV, the most severe contingency scenario. In this scenario, the integrated PWGNs respond dynamically, adjusting pump electricity consumption and water flow rates to manage the disruptions. Notably, during peak demand hours (e.g., 6 PM to 10 PM), pump electricity consumption reaches a maximum of approximately 18 MW, highlighting the heavy reliance of water networks on electricity for pumping operations. Conversely, in low-demand periods, pump electricity consumption drops to around 4 MW, demonstrating the system's flexibility in conserving energy when possible. The water tank flow rate also fluctuates significantly, ranging from $-2000 \text{ m}^3/\text{h}$ (indicating water outflow to meet demand) to $+3000 \text{ m}^3/\text{h}$ (indicating water inflow for storage replenishment). These adjustments help stabilize water availability during peak electricity demand hours by balancing

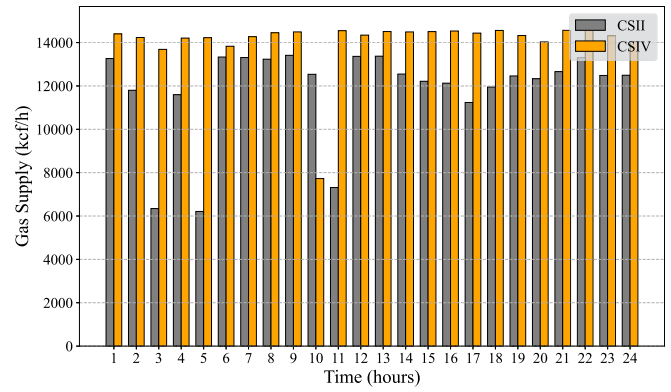


Fig. 13. Total gas supply in CSII and CSIV: IEEE 118-bus test system.

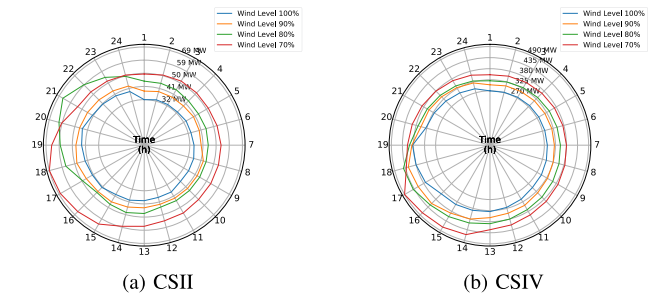


Fig. 14. Total load shedding distribution in (a) CSII and (b) CSIV under varying wind penetration levels: IEEE 118-bus test system.

consumption with reserves. This interdependent response enhances resilience by ensuring that water supply remains consistent even under extreme operational conditions in the power and gas networks. Fig. 13 compares the total gas supply under contingency scenarios CSII and CSIV within the 20-node Belgian system configuration. In CSII, the gas supply remains relatively stable, averaging around 12,000 kcf/h with only minor fluctuations. In contrast, in CSIV the gas supply shows more pronounced adjustments, particularly during high-demand periods. For instance, at peak hours (e.g., 6 PM to 10 PM), the gas supply increases to approximately 14,500 kcf/h in CSIV to compensate for the pipeline disruption, compared to an average of 12,500 kcf/h in CSII. This adjustment represents an additional 15% supply increase to meet demand under compounded disruptions. Fig. 14 illustrates the load shedding profiles across contingency cases CSII and CSIV, highlighting the variations in shedding patterns under different wind penetration levels and contingency scenarios. In CSII, Fig. 14(a), where a single line failure disrupts power flow, load shedding intensifies at lower wind levels. For instance, the total load shedding reaches its highest at 70% wind penetration, particularly between 6 PM to 10 PM, aligning with peak demand hours. This demonstrates how the system becomes more reliant on conventional generation sources as wind output decreases, leading to significant load shedding during critical hours. Conversely, in CSIV (Fig. 10(b)), which involves simultaneous failures in both power lines and gas pipelines, the load shedding is markedly higher due to the compounded effect of these failures. The total load shedding peaks at 359 MW during

TABLE V
PERFORMANCE OF INTEGRATED MODEL UNDER CONTINGENCY SCENARIOS:
IEEE-118-BUS TEST SYSTEM

Case study	CSI	CSII	CSIII	CSIV
Water Demand Level (%)	100%	110%	120%	130%
Total Cost (k\$)	3981.7	4396.6	4177.9	4580.5
Load Shedding (MW)	0	51.3	29.7	278.2
Cost Increase (%)	0%	10.4%	4.9%	15%
Computation Time (s)	1217	1449	1255	1500

70% wind penetration, with a clear escalation in shedding levels during the evening peak hours. Even at higher wind levels, the combined failures impose significant stress on the system, as evidenced by elevated shedding levels compared to CSII.

F. Performance Comparison With the State-of-The-Art Models

The proposed framework was evaluated against state-of-the-art methods using a modified IEEE 118-bus test system interconnected with five water networks and a 20-node gas network. The water networks were connected at buses 1, 27, 39, 74, and 102, while wind generation units were located at buses 8, 33, and 75. This setup was tested under varying water demand scenarios, ranging from 100% to 130% of baseline levels, to assess the framework's ability to handle interdependent operations in PWGNs.

Table V presents the results of the four case studies (CSI to CSIV). In CSI, representing normal operation with 100% water demand, the system incurs a total cost of \$3981.7k, with no load shedding observed. This serves as the baseline for comparison. As contingencies are introduced, such as a power line outage in CSII (110% water demand) or a gas pipeline failure in CSIII (120% water demand), the system experiences higher operational costs and load shedding. In CSII, the total cost rises to \$4396.6k, with a load shedding of 51.3 MW. The slight increase in water demand coupled with the power line outage primarily drives this cost increase of approximately 10.4% from CSI. On the other hand, CSIII results in a lower load shedding of 29.7 MW, but the total cost remains significant at \$4177.9k due to the compounded impact of gas network constraints. This case demonstrates the framework's ability to balance inter-network interdependencies by reallocating resources to mitigate the effects of individual contingencies. CSIV, representing the most severe scenario with simultaneous power and gas network failures and 130% water demand, leads to the highest total cost of \$4580.5k and a substantial load shedding of 278.2 MW. This corresponds to a 15.0% increase in cost compared to CSI, highlighting the compounded effects of multi-network disruptions under high water demand conditions. Such results highlights the critical need for joint optimization in enhancing system resilience and minimizing cascading failures. The computational performance of the framework remains robust across all scenarios, with solution times ranging from 1217 seconds in CSI to 1500 seconds in CSIV. For CSIV, the model was stopped at 1500 seconds, and the results were taken at that point to

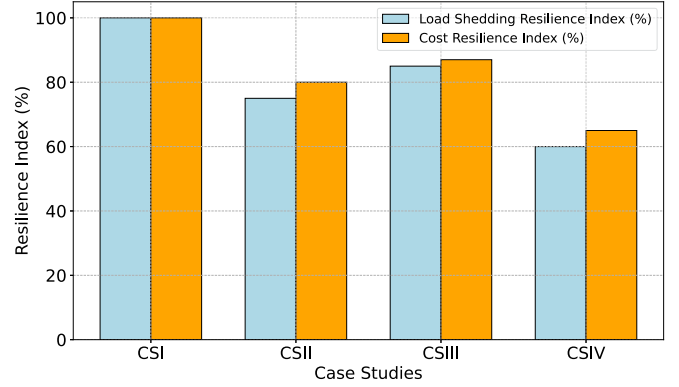


Fig. 15. Resilience indices for cost and load shedding across case studies: IEEE 118-bus test system.

ensure practical computational limits. To further evaluate the framework's performance, resilience indices for cost and load shedding were calculated and are presented in Fig. 15. Under CSI, resilience indices are at 100% for both metrics, as no contingencies occur. As the scenarios increase in severity (e.g., CSIII and CSIV), the proposed framework sustains significantly higher resilience indices compared to conventional bi-network models. For instance, under CSIV, the *Load Shedding Resilience Index* and *Cost Resilience Index* remain above 65% and 60%, respectively, compared to indices below 40% in similar studies on bi-network optimization under extreme conditions [12] and [37]. These findings align with prior multi-network studies, such as [23] and [25], validating the proposed model's advantages in enhancing resilience and reducing operational costs under contingency scenarios, while effectively addressing interdependencies across multi-energy systems.

V. CONCLUSION

This study proposed a novel operational strategy for interconnected PWGNs under emergency conditions, effectively addressing system interdependencies through efficient linearization techniques. By transforming a mixed-integer nonlinear programming (MINLP) model into an equivalent mixed-integer linear programming (MILP) formulation, the framework could reduce operational costs and minimize load shedding while maintaining system resilience with an affordable computational resources. The results across multiple case studies and on two different-size interconnected networks validate the scalability and robustness of the proposed approach, providing a practical and efficient solution for joint PWGN operation under complex contingency scenarios. By leveraging interdependencies among infrastructures, the model enhances resilience and cost efficiency, establishing a benchmark for coordinated multi-energy system optimization.

Future research will focus on enhancing the framework's applicability to real-time operations and larger-scale systems while addressing uncertainties in load profiles, renewable generation, and extreme weather events. Expanding the model to incorporate multiple system operators and advanced linearization

techniques will further improve its accuracy and computational efficiency. These advancements aim to develop a comprehensive framework for real-time, integrated operation of multi-energy systems, with the goal to enhance the resilience and reliability in interconnected infrastructures.

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